

Multiple Choice (10 marks)

1. What Mars mission will be landing on May 25 2008 and will dig a trench into (hopefully) ice-rich soil?
 - (a) ExoMars
 - (b) Mars Exploration Rovers
 - (c) Mars Science Laboratory
 - (d) Phoenix Mars Lander
2. The term Schwarzschild radius usually describes properties of ...
 - (a) red dwarfs.
 - (b) pulsars.
 - (c) black holes.
 - (d) galaxies.
3. How do astronomers think Jupiter generates its internal heat?
 - (a) nuclear fusion in the core
 - (b) by contracting changing gravitational potential energy into thermal energy
 - (c) internal friction due to its high rotation rate
 - (d) chemical processes
4. The nebular theory of the formation of the solar system successfully predicts all but one of the following. Which one does the theory not predict?
 - (a) Planets orbit around the Sun in nearly circular orbits in a flattened disk.
 - (b) the equal number of terrestrial and jovian planets
 - (c) the craters on the Moon
 - (d) asteroids Kuiper-belt comets and the Oort cloud
5. The famous Drake equation attempts to answer the following question:
 - (a) Will the Sun become a black hole?
 - (b) Is the universe infinitely large?
 - (c) How old is the visible universe?
 - (d) Are we alone in the universe?

True / False (10 marks)

Answer True or False

6. True or False: When a blackbody is cooled to half its original temperature, the power emitted decreases to 1/16 of its original value and the peak wavelength doubles.
7. True or False: Knowing an object's actual brightness and its apparent brightness allows for the estimation of its distance from the observer.
8. True or False: Planet X is tidally locked, resulting in the absence of a solar day as it does not complete a rotation relative to its sun.
9. True or False: Bolometric luminosity refers specifically to the luminosity integrated over visible wavelengths only.
10. True or False: Asteroids emit a significant amount of their own radiation, which makes them stand out in sky surveys.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the major hypotheses regarding the origin of the Moon, and critically analyze why the tidal forces leading to a split from Earth is not a widely accepted explanation.
12. Discuss the significance of Betelgeuse within the Orion constellation, including its characteristics, cultural impact, and role in stellar evolution studies.

End of Paper